

Contrastive Domain Adaptation for Retinal Image Quality Assessment

(2024)

Problem

Assessing retinal fundus image quality automatically across different camera manufacturers.

Method

A contrastive domain adaptation method trained to be invariant to camera-specific artifacts.

Dataset

EyeQ retinal image quality dataset.

Evaluation

Quality classification accuracy measured across camera domains.

Results

Cross-domain quality classification accuracy improved from 74% to 89% after domain adaptation.

Limitations

Quality assessment does not yet feed back into downstream diagnostic accuracy measurement.